ASSESSMENT SOLUTION FOR DELIVERABLE 1

COURSE: MSC COMPUTER SCIENCE
MODULE: SOFTWARE ANALYSIS AND DESIGN
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CASE STUDY: INTERACTIVE E-LEARNING SYSTEM

E-learning management system (e-LMS) facilitates the students, teachers and academic institution to learn and deliver education to the students through online learning mediums and applications. The online learning management system supports online registration process such as download lecture notes, upload assignment, attend online test, view videos and attend online lecture. Similarly, lecturer can deliver lectures and presentations, do group discussions, receive student queries, upload module contents, etc.

During the COVID19 pandemic, several services were shifted to be provided online in different industries such as healthcare, education, food delivery and others. The schools, colleges and universities are reaching out students to deliver online learning facilities using internet and technical infrastructure to provide educational opportunities at distant places with effective learning mediums on minimal cost. E-learning systems provide opportunity for regional and rural students to learn and complete their education from home or distant places. In the UK, all universities moved to the distance/online learning to conduct all teaching, learning, research and management activities.

This system will serve students, lecturers and admins to provide an interactive e-learning management system services. It should provide extra and different functionalities comparing to the existing systems, considering the below prospective**.**

Student Prospective

Student experience determines the usability, efficiency, insight information, user interface rating for a e-LMS. It should enable students to enroll to modules, download materials, attend online meetings, ask enquiry, submit assessments, conduct online quizzes and tests, ... etc.

Lecturer Prospective

Lecturer uses the e-LMS as an information delivery portal/tool; they would download the submitted assignments, add test and quiz, add assignment and upload lecture material on the portal. Lecturer can create online meetings and invite students to attend these meetings. They also can find statistics about the performance of the students in each module.

Admin Prospective

Admin view the e-LMS in terms of security and reliability. The system should be secure, reliable and support large set of administrative functions that make the admin operation smooth and easy. They can find statistics about the main activities of all modules.

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1.1.1 Phases of the SE life cycle

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1.1.1.2 System Design

1.1.1.3 Implementation

1.1.1.4 Testing

1.1.1.5 Deployment

1.1.1.6 Maintenance and Support

1.2 Question 2 - Identifying the different SE methodologies and describing how to select an appropriate methodology for a project

Here are seven different methodologies:

1.2.1 Waterfall Model

Advantages of the Waterfall Model include:

Disadvantages of the Waterfall Model include:

1.2.2 Agile Methodology

1.2.2.1 Types of Agile methodology

Rapid Application Development (RAD)

Advantages of RAD include:

Disadvantages of RAD include:

1.2.3 Spiral Model

Advantages of the Spiral Model include:

Disadvantages of the Spiral Model include:

1.2.4 Lean Software Development

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dvantages of Lean Software Development include:

Disadvantages of Lean Software Development include:

1.2.5 V-Model

Advantages of the V-Model include:

Disadvantages of the V-Model include:

1.2.6 Prototyping Model

Advantages of the Prototyping Model include:

Disadvantages of the Prototyping Model include:

Question 4 - Main ethical issues that may be faced after the analysis and design conduction

Table 2.4.1: Table for ethical and legal (EL) issues

Identifier	Description	Justification
EL1	Data Privacy and Protection	Ensuring that user data (students, lecturers, and admins) is securely stored and accessed only by authorised individuals. Compliance with data protection regulations like GDPR is essential.
EL2	Accessibility	Designing the e-LMS to be accessible for all users, including those with disabilities, ensuring equal opportunities for everyone to access and benefit from the system.
EL3	Intellectual Property Rights	Properly handling copyrighted materials, such as lecture notes, videos, and other educational resources, to avoid legal issues and respect the rights of content creators.
EL4	Digital Divide	Acknowledging that not all users may have equal access to the internet, devices, or technical skills, which can lead to inequality in educational opportunities.
EL5	Content Quality and Accuracy	Ensuring that the educational materials provided within the e-LMS are accurate, up-to-date, and of high quality, to maintain the integrity and credibility of the system.

EL6	Fair Assessment and Evaluation	Implementing transparent and unbiased assessment and evaluation mechanisms, to ensure that all students are treated fairly and have equal opportunities to succeed.
EL7	User Consent and Opt-out Options	Obtaining explicit user consent for data collection, processing, and sharing, and providing clear opt-out options for users who do not wish to share their data.
EL8	Online Security and Protection from Cyber Threats	Protecting the e-LMS from cyberthreats, such as hacking, data breaches, and DDoS attacks, to maintain user trust and ensure the system's reliability and security.
EL9	Transparency	Being transparent about how the e-LMS operates, collects data, and makes decisions, to establish trust and maintain open communication with users.
EL10	Compliance with Local and International Laws	Ensuring that the e-LMS complies with all applicable local and international laws and regulations, such as copyright laws, data protection regulations, and educational standards.

EL11	Avoiding Bias in Algorithms and System Design	Ensuring that algorithms used in the e-LMS, such as recommendation systems or grading, do not introduce or reinforce bias against any group.
EL12	Cultural Sensitivity and Inclusivity	Designing the e-LMS to be culturally sensitive and inclusive, taking into account diverse backgrounds, perspectives, and learning styles.

EL13	Protection of Minors	Implementing proper measures to protect minors from harmful content, cyberbullying, and other online threats within the e-LMS environment.
EL14	Responsible Use of Third-Party Tools and Services	Ensuring that any third-party tools or services integrated into the e-LMS adhere to the same ethical and legal standards as the e-LMS itself.
EL15	Environmental Impact	Minimising the environmental impact of the e-LMS, such as energy consumption and electronic waste, by promoting sustainable design practices.

PART 2

2.1 Question 1 - Appropriate methodology for analysing and designing the system

2.2 Question 2 – Tools and Activities used in the requirement phase of the project

2.3 Question 3 - Foldable Concept Map (FCM) Model

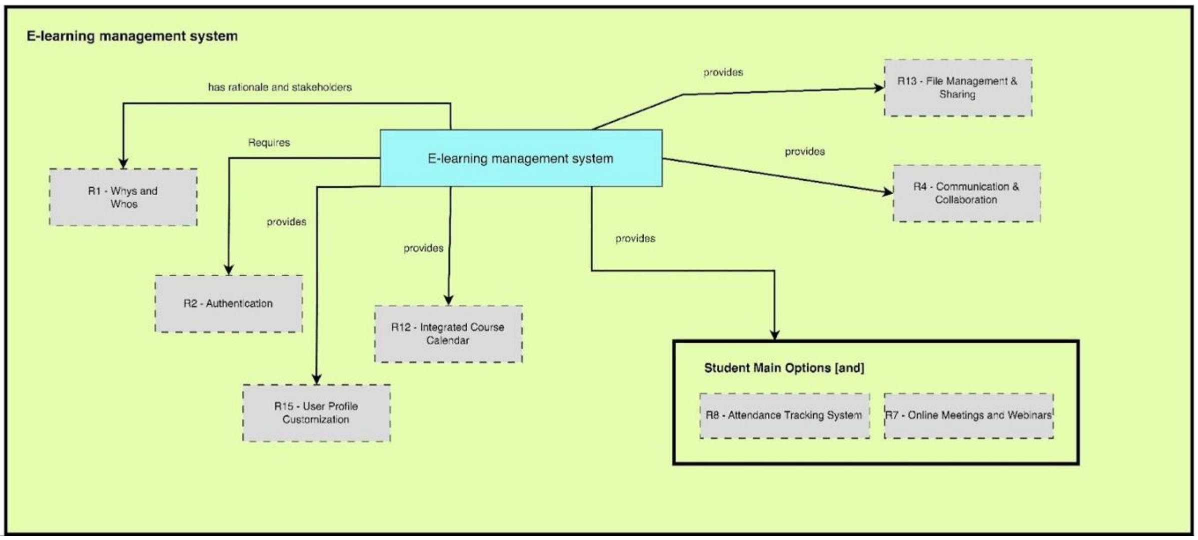


Figure 9: *FCM for e-LMS Student main options.*

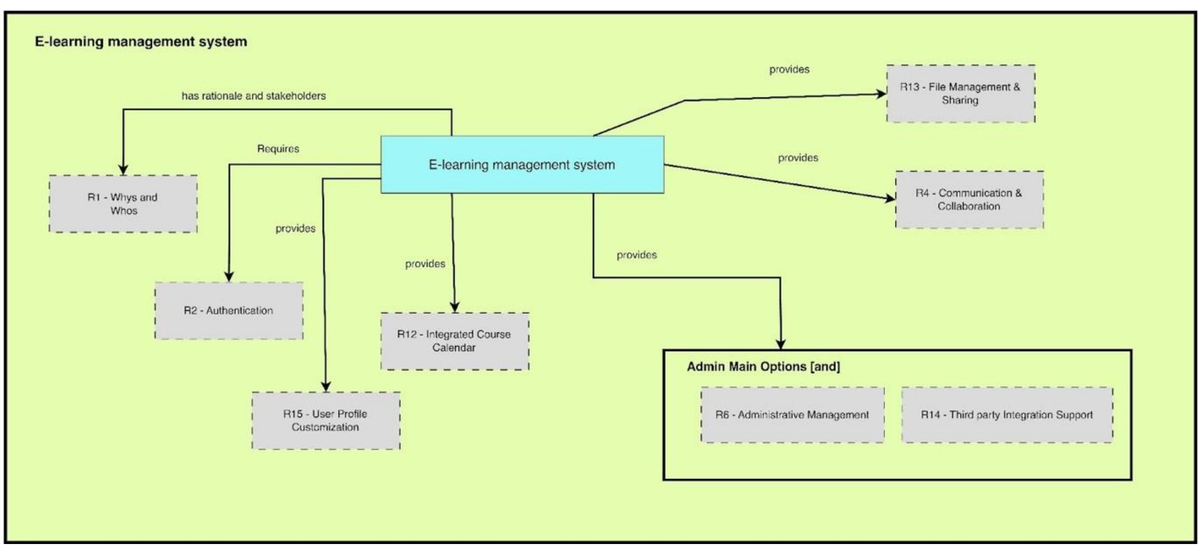


Figure 10: *FCM for e-LMS Admin main options.*

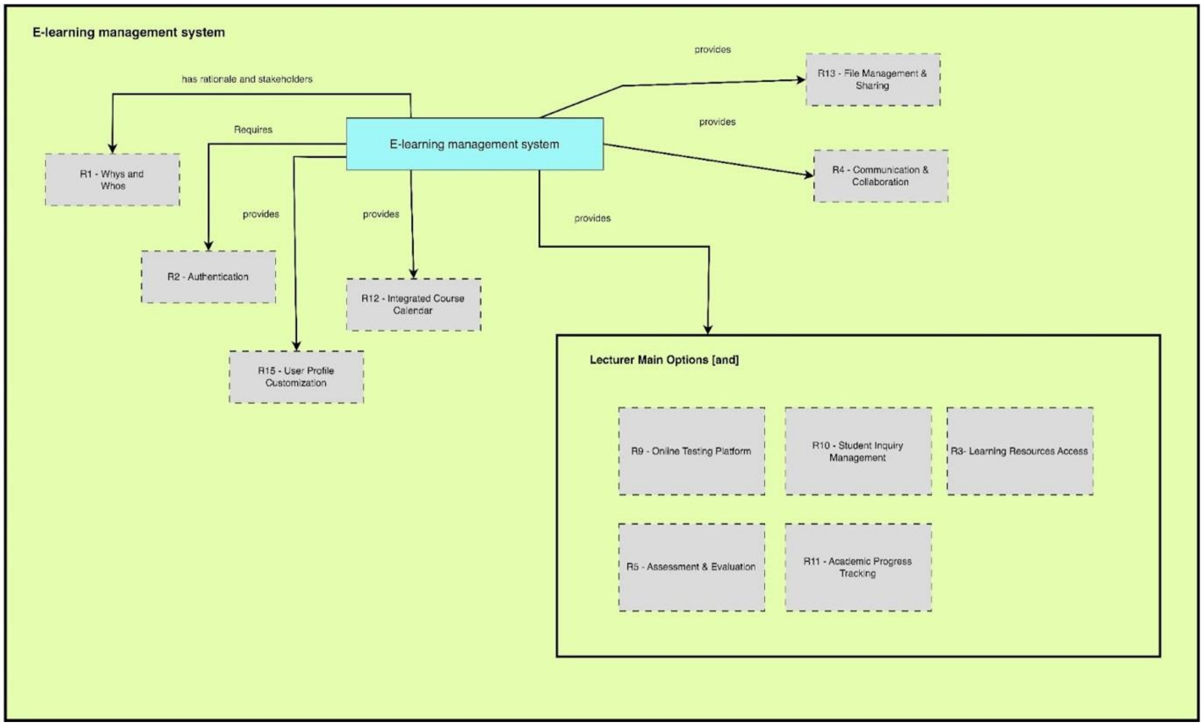


Figure 11: *FCM for e-LMS Lecturer main options.*

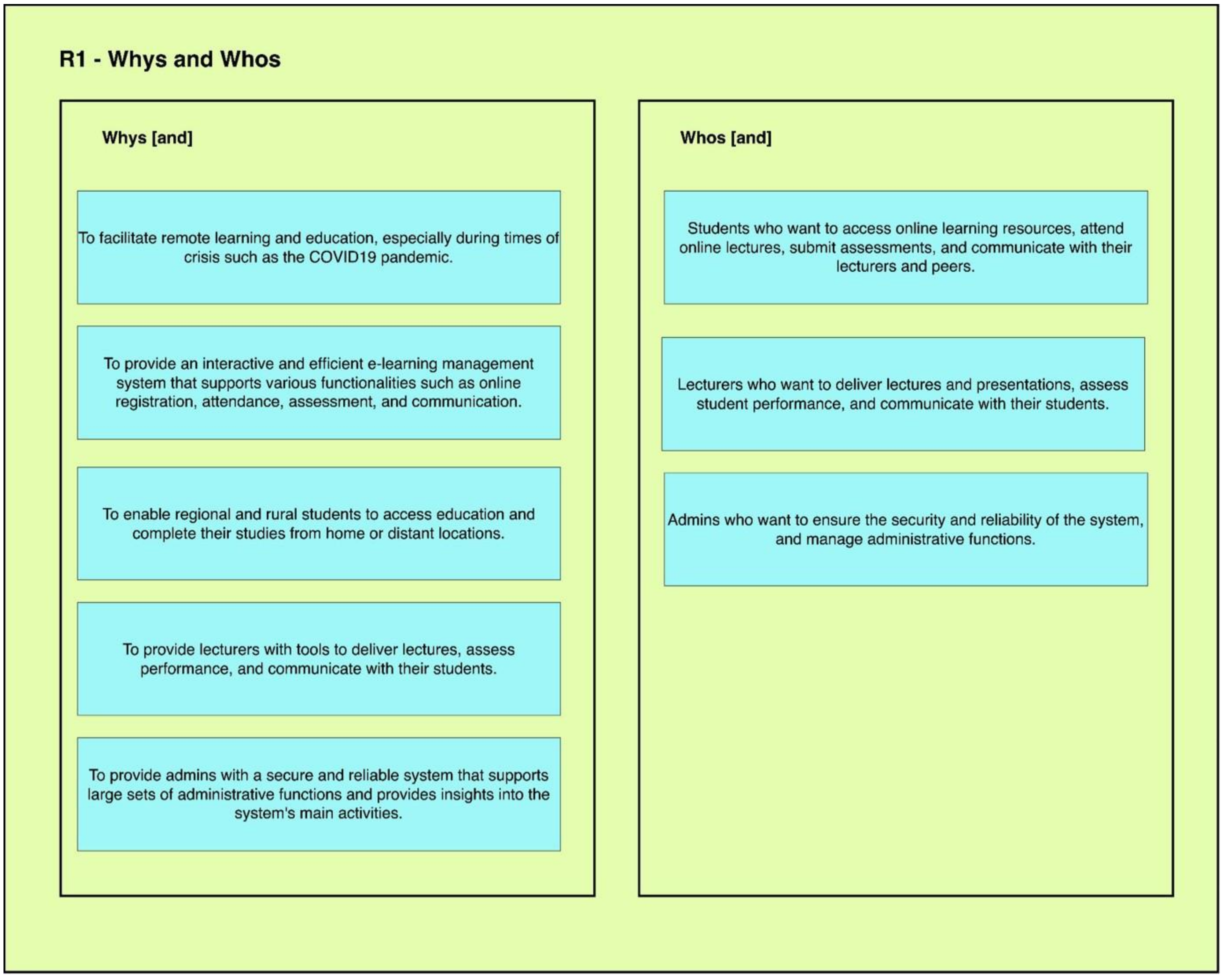


Figure 12: *FCM for e-LMS R1 Whys and Whos*

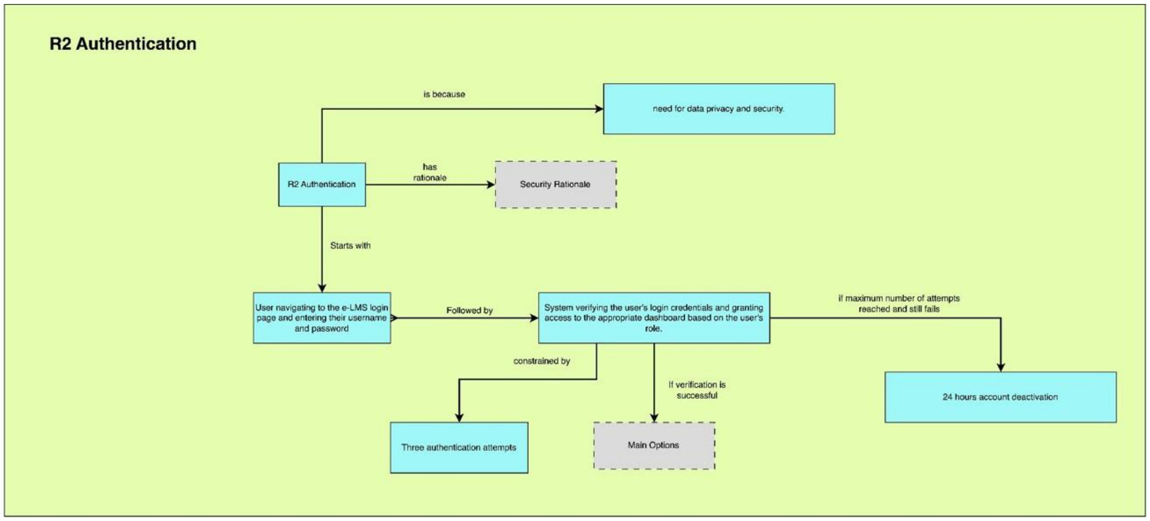
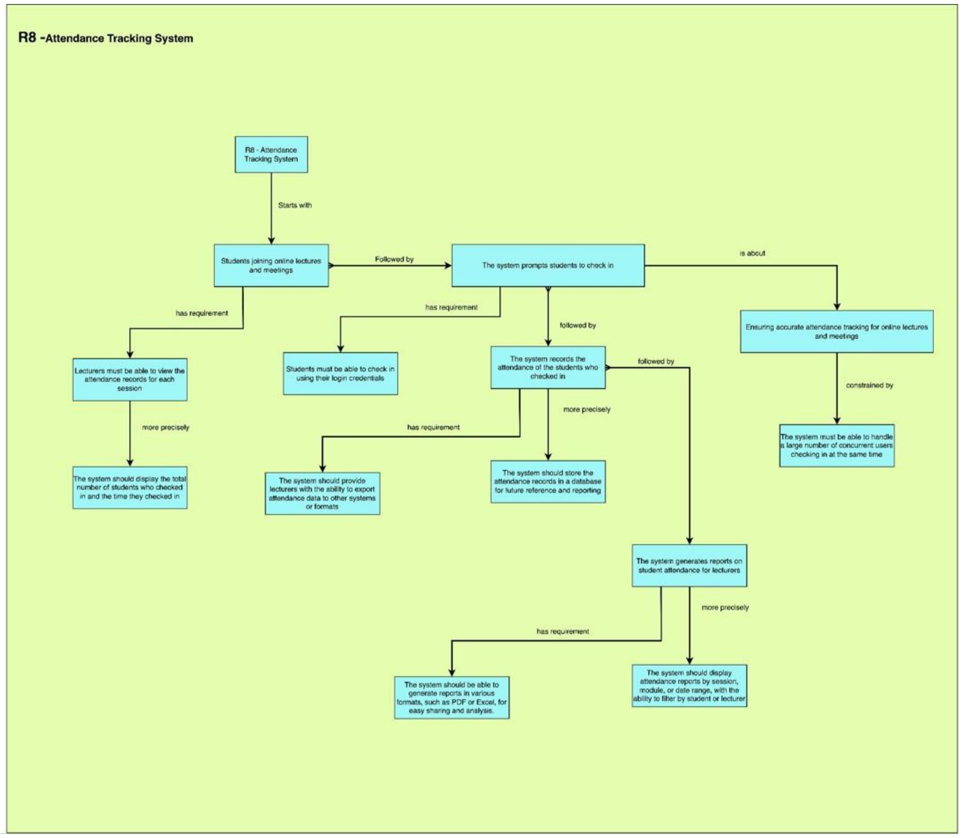


Figure 13: *FCM for e-LMS Proxy R2 Authentication.*



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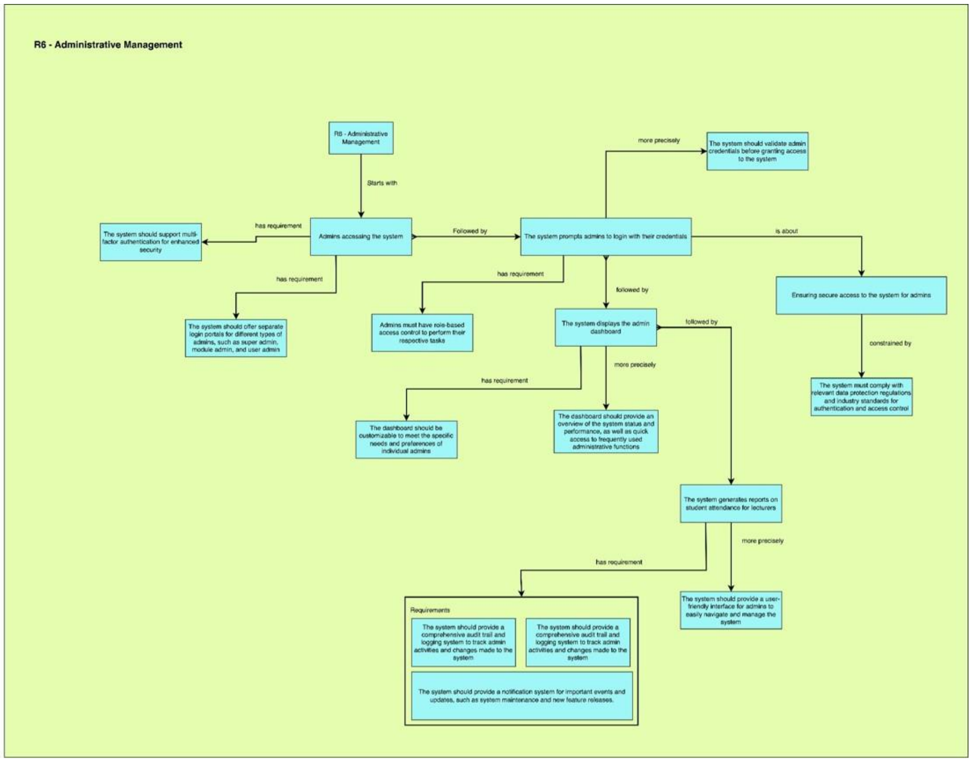

Figure 15: *FCM for e-LMS Proxy R6-Administrative Management*

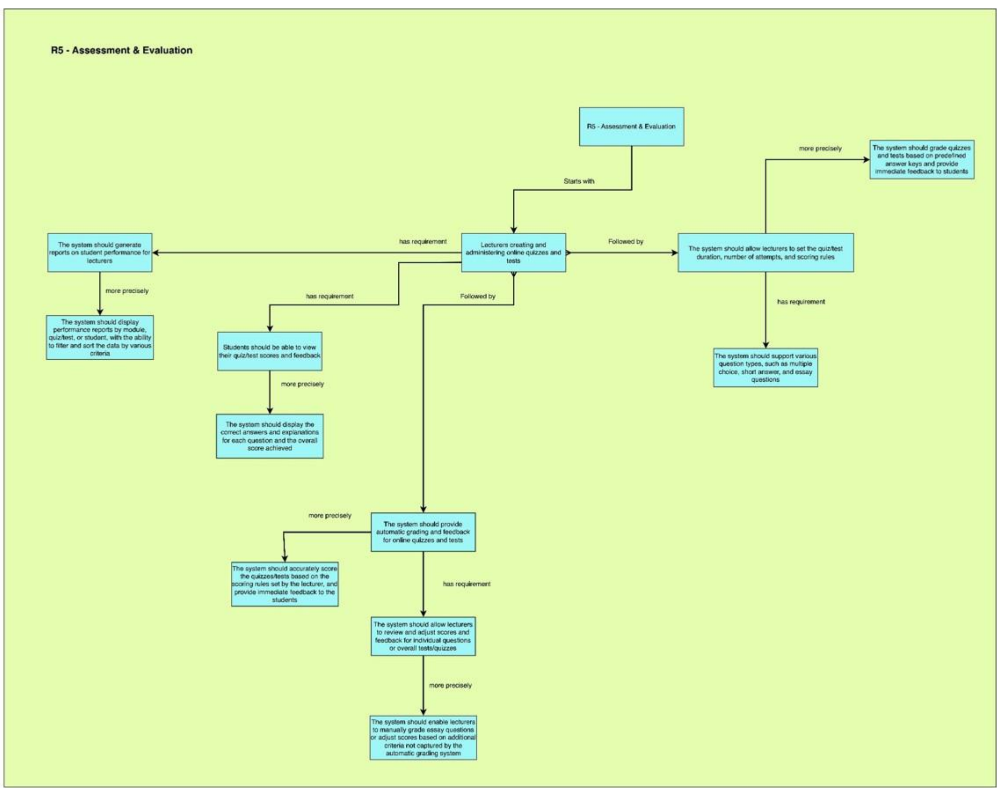

Figure 16: *FCM for e-LMS Proxy R5-Assessment and Evaluation*

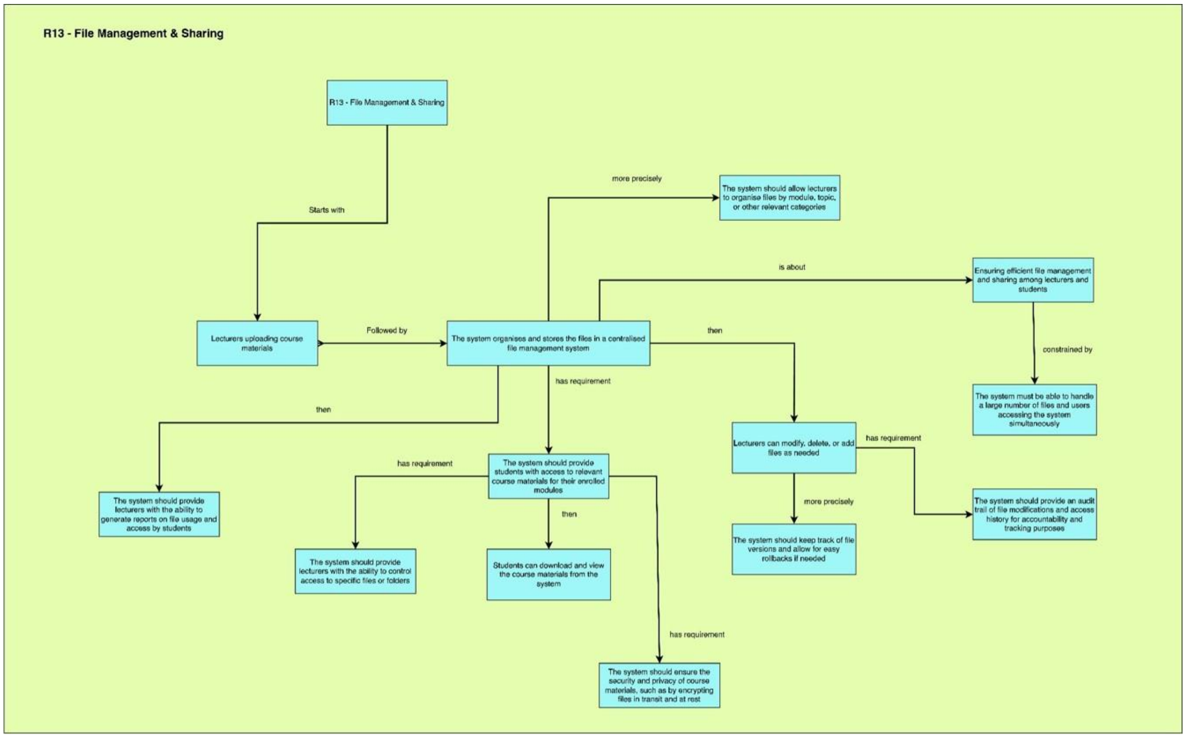

Figure 17: *FCM for e-LMS Proxy R13-File Management & Sharing*

2.4 Question 4 – Functional and Non-Functional Requirements Table

Functional Requirement Table:

R1 The e-LMS shall provide a user-friendly interface for students, lecturers, and admins to access and manage their respective tasks and responsibilities.

R1.1 The system shall offer separate login portals for students, lecturers, and admins with role-based access control.

R1.2 The system shall support multiple device compatibilities, such as computers, tablets, and smartphones.

R2 The e-LMS shall support registration and enrollment processes for students.

R2.1 The system shall allow students to search and enrol in available modules.

R2.2 The system shall allow students to view their enrolled modules and track their progress.

R2.3 The system shall enable students to download module materials, such as lecture notes and presentations.

R3 The e-LMS shall facilitate access to multimedia learning resources for students.

R3.1 The system shall allow students to view and download recorded lecture videos.

R3.2 The system shall provide access to additional resources, such as articles and supplementary materials.

R4 The e-LMS shall support communication and collaboration among students and lecturers.

R4.1 The system shall provide a messaging feature for students to ask questions and receive feedback from lecturers.

R4.2 The system shall enable group discussions and forums for collaborative learning.

R5 The e-LMS shall provide tools for assessment and evaluation.

R5.1 The system shall allow students to submit assignments and lecturers to download and grade them.

R5.2 The system shall support the creation and administration of online quizzes and tests.

R5.3 The system shall generate reports on student performance for lecturers.

R6 The e-LMS shall offer administrative tools for managing modules and users.

R6.1 The system shall allow admins to create, modify, and delete modules.

R6.2 The system shall provide tools for managing student and lecturer accounts.

R7 The e-LMS shall support online meetings and webinars.

R7.1 The system shall allow lecturers to create online meetings and invite students to attend.

R7.2 The system shall provide tools for recording and archiving online meetings for later access.

R8 The e-LMS shall facilitate attendance tracking for online lectures and meetings.

R8.1 The system shall allow students to check in to online lectures and meetings.

R8.2 The system shall provide reports on student attendance for lecturers.

R9 The e-LMS shall enable students to complete online quizzes and tests.

R9.1 The system shall allow lecturers to create and manage quizzes and tests.

R9.2 The system shall provide automatic grading and feedback for online quizzes and tests.

R10 The e-LMS shall provide a platform for students to submit and track inquiries.

R10.1 The system shall allow students to submit inquiries to lecturers and support staff.

R10.2 The system shall enable students to track the status of their inquiries.

R11 The e-LMS shall provide a dashboard for students to track their overall academic progress.

R11.1 The system shall display a summary of module completion, grades, and attendance for each student.

R11.2 The system shall enable students to set goals and track their progress toward achieving them.

R12 The e-LMS shall offer an integrated calendar for managing deadlines and events.

R12.1 The system shall display important dates, such as assignment deadlines, test dates, and online meeting schedules.

R12.2 The system shall enable students and lecturers to set reminders and notifications for upcoming events.

R13 The e-LMS shall provide an integrated file management system for organising and sharing course materials.

R13.1 The system shall enable lecturers to upload, organise, and share files with their students.

R13.2 The system shall allow students to access and download files for their enrolled modules.

R14 The e-LMS shall support the integration of third-party tools and resources.

R14.1 The system shall enable the integration of external educational resources, such as e-books and research databases.

R14.2 The system shall provide support for connecting with third-party communication and collaboration tools.

R15 The e-LMS shall offer customizable user profiles for students, lecturers, and admins.

R15.1 The system shall enable users to personalise their profiles with relevant information, such as contact details, educational background, and areas of expertise.

R15.2 The system shall provide options for users to adjust their account settings and notification preferences.

Non-Functional Requirement Table:

N1 The e-LMS shall provide a response time of fewer than 6 seconds for 95% of user actions.

N2 The system shall support a minimum of 1,000 concurrent users without performance degradation.

N3 The e-LMS shall maintain a minimum uptime of 99.5%.

N4 The system shall ensure data privacy and security in compliance with relevant data protection regulations.

N5 The e-LMS shall provide a user-friendly interface, with a minimum average user satisfaction rating of 4 out of 5.

N6 The e-LMS shall provide a high level of system availability during peak usage hours, with a minimum uptime of 99.9% during these times.

N7 The system shall provide efficient and effective load balancing and scalability capabilities to handle increased traffic during peak usage hours.

N8 The system shall provide a high level of data security, utilising encryption for data in transit and at rest, and regularly conducting security audits to ensure compliance with relevant security standards.

N9 The e-LMS shall support integration with a range of internet browsers, ensuring compatibility with popular browsers such as Google Chrome, Mozilla Firefox, and Safari.

N10 The system shall provide a comprehensive helpdesk and support service to address user issues and inquiries promptly, with a maximum response time of 24 hours for non-critical issues.

N11 The system shall deactivate a user account for a period of 24 hours after three consecutive failed login attempts, to prevent unauthorised access and improve security.

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